Pavan Prudhvi Madhamsetty

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Education

University of Florida — Gainesville, USA
Master of Science in Computer Science; GPA: 3.8/4.0 — *Aug 2024 – May 2026*

Vellore Institute of Technology — Tamil Nadu, India
Bachelor of Technology in Computer Science and Engineering — *Jul 2019 – Aug 2023*

Technical Skills

Languages: Python, Java, SQL, TypeScript, C++
Backend & Systems: Flask, Spring Boot, REST APIs, webhooks, Next.js, gRPC, Raft
AI & Data: LangChain, Azure OpenAI, RAG, vector embeddings, XGBoost, scikit-learn, pandas, NumPy, Power BI
Databases, Cloud & Testing: MySQL, SingleStore, PostgreSQL, AWS EC2, Azure, pytest, Postman, Google Test, Git

Experience

Dhurin Data Sciences Pvt Ltd — Noida, India
Software Engineer — *Jan 2023 – Jul 2024*

- Developed a LangChain ReAct agent on GPT-3.5 that turns plain-English questions into SQL queries against enterprise data for multiple enterprise clients, following the ReAct reasoning-and-acting pattern.
- Shipped a retrieval-augmented chatbot on Azure OpenAI that grounds GPT responses in context retrieved through vector embeddings for client-facing support and question answering.
- Wrote the Flask backend for a multi-client conversational platform, with one unified webhook handler integrating the Twilio WhatsApp and Slack APIs for several client organizations.
- Processed large banking transaction datasets with pandas and NumPy and trained XGBoost and Random Forest credit risk models for a banking client, using SMOTE to correct class imbalance.

Mavenir Systems, Inc. — Bangalore, India
Software Engineer — *May 2021 – Jul 2022*

- Built Spring Boot REST APIs in Java on AWS EC2 to handle 5,000+ concurrent sessions, and set up a pytest and Postman testing framework for unit and API tests that reached 80% code coverage.
- Optimized MySQL query performance by designing composite indexes for frequent query patterns, cutting query time by roughly 30%, and migrated high-throughput tables to SingleStore.
- Implemented two outlier detectors with Seldon Alibi Detect for service monitoring: a variational autoencoder (VAE) trained on CIFAR-10 images and a Spectral Residual time-series model.
- Evaluated both detectors with F1 scores and confusion matrices, tuning detection thresholds to reduce false-positive alerts, and monitored the deployed services through Azure Application Insights.

Projects

Turnout Planner | *Next.js, TypeScript, Prisma, PostgreSQL* — Fall 2025

- Built and deployed a live operations tool for barns in Next.js and TypeScript, where recurring group schedules auto-generate daily turnout events, with drag-and-drop reassignment, bulk actions and an overdue-reminders bell.
- Modeled the data in Prisma on PostgreSQL, routed every mutation through Server Actions, used ON DELETE RESTRICT foreign keys to block orphaned records, and pinned dates to UTC for consistent rendering.

Distributed Key-Value Store | *C++, Raft, gRPC, LSM-Tree* — Spring 2025

- Built a sharded key-value store in C++ that partitions keys with consistent hashing and replicates each shard through Raft consensus, with gRPC and Protocol Buffers for communication between nodes.
- Wrote an LSM-tree storage engine and a thread-pool request layer synchronized with mutexes, and verified behavior under network partitions with Google Test suites built through CMake.